Thirty years ago, there was hardly any private-sector power-generation capacity in India. Power generation was opened for private-sector investment in 1991. There were many policy problems in the initial years. Only 1430 MW of power-generation capacity was added in the private sector in the Eighth Plan (1992–1997), more than half of which was in the gas power plant segment (762 MW). Power generation in the private sector started assuming notable proportions by the end of the 1990s. At the end of financial year 2011–12, India had a total power-generation capacity of 2.11 lakh MW—the private sector's share was 29.11 per cent with 61,409 MW of capacity. Very soon, this share will exceed 50 per cent as most new generation is coming up in the private sector.

India's renewables-based power-generation capacity (primarily hydro, solar and wind) at the end of October 2020 was 1.35 lakh MW, comprising hydro (45,699 MW) and renewables (89,636 MW), which made up over 36 per cent of total installed capacity. At the turn of the millennium, solar and wind power-generation capacity was quite insignificant. Today, the largest addition every year is from the renewables segment.

The only renewable power generation taking place in India in 1990 was hydropower. The country had total hydro and renewable power generation-capacity of 18,307 MW at the end of financial year 1990. The first wind power-generation capacity started coming on stream in 1993–94 and India's installed renewable power (other than hydro) grew to 902 MW by financial year 1997. This then grew, primarily driven by wind power generation, to 7760 MW by 2007. Once solar power became competitive (initially with government subsidy support and after 2015 without such support), renewable power generation (other than hydro) really took off. India's renewable power capacity (other than hydro) was 24,503 MW by the end of 2012; it reached about 87,000 MW by the end of financial year 2020.

From 2017 to 2020, while thermal power-generation capacity from all sources (coal and gas) grew by only about 12,000 MW, renewable power